Analytical Chemistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

2.2 Let Titration Begin

2.2 Titration

Overview

2.2-1 Titration curve

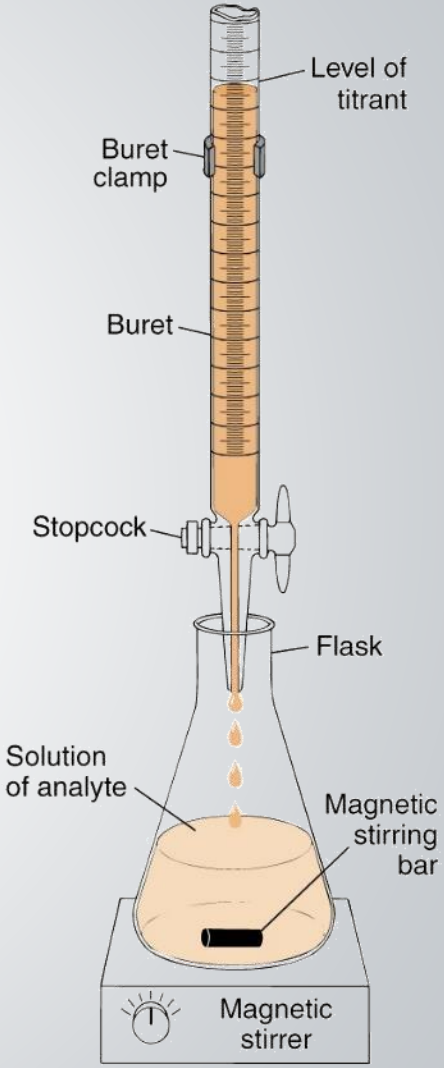
2.2-2 Titration error

2.2-3 Find the end point

2.2-4 Titration types

2.2-4 Titration types

- Acid-base titration
- Precipitation titration
- Redox- titration



Typical setup for titration

❑Precipitation titration

Precipitation titration

Solubility product (K_{sp})

Potentiometric titration

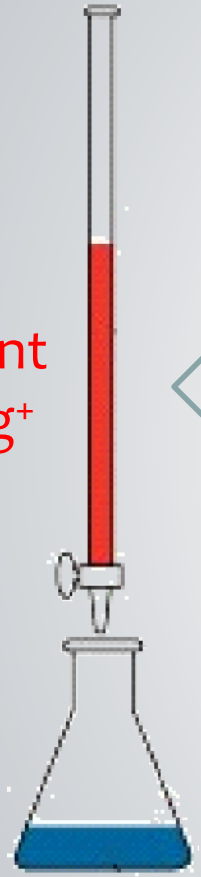
Mohr titration

Volhard titration

Fajans titration

❑ Precipitation titration

Titration based on the reaction in which the product of precipitates



Titrant of Ag^+

$$Ag^+ + I^- \overset{K_{sp}}{\rightleftharpoons} AgI\downarrow$$

Solubility product (K_{sp})

$$K_{sp} = [Ag^+][I^-]$$

$C_{Ag^+} = [Ag^+]$

v : Titrated volume

$[I^-] = C_I$

Vol. = V_0

Question:

Find $[Ag^+]$ as a function of the volume V added to the solution?

Ans.: first calculate $[I^-]$, then find $[Ag^+]$

$$[Ag^+] = K_{sp} / [I^-]$$

Initial I^- moles

I^- moles that reacted

$[I^-]$ from dissolution of AgI precipitate, Which equals to $[Ag^+]$

$$[I^-] = \frac{\overbrace{V_0 C_{I^-}} - \overbrace{v C_{Ag^+}}}{V_0 + v} + \frac{Ksp}{[I^-]}$$

$[Ag^+]$ in solution, determined by added Ag^+ that did not precipitate. For each Ag^+ ion there is an I^- ion.

$$[I^-] = \frac{V_0 C_{I^-} - vC_{Ag^+}}{V_0 + v} + \frac{Ksp}{[I^-]}$$

Extracting I^- (quadratic eq.):

$$[I^-] = \frac{V_0 C_{I^-} - vC_{Ag^+}}{2(V_0 + v)} + \sqrt{\left[\frac{V_0 C_{I^-} - vC_{Ag^+}}{2(V_0 + v)}\right]^2 + Ksp}$$

Finding $[Ag^+]$:

$$[Ag^+] = \frac{Ksp}{\frac{V_0 C_{I^-} - vC_{Ag^+}}{2(V_0 + v)} + \sqrt{\left[\frac{V_0 C_{I^-} - vC_{Ag^+}}{2(V_0 + v)}\right]^2 + Ksp}}$$

Precipitate titration curve

*Three regions for titration curve

K_{sp} (AgI) = 8.3×10^{-17}

1. Before equivalence point, 0< V <V_e

$p[Ag^+]_f$ is determined by AgI redissolution

$[Ag^+]_f = K_{sp}/[I^-]$ $\quad [I^-]_f = \frac{[I^-]V_o - [Ag^+]v}{V_o + v}$

$p[Ag^+]_f = 14.84 \ (V = 10\ mL)$

2. At equivalence point, V = V_e

$p[Ag^+]_f$ is determined by K_{sp}

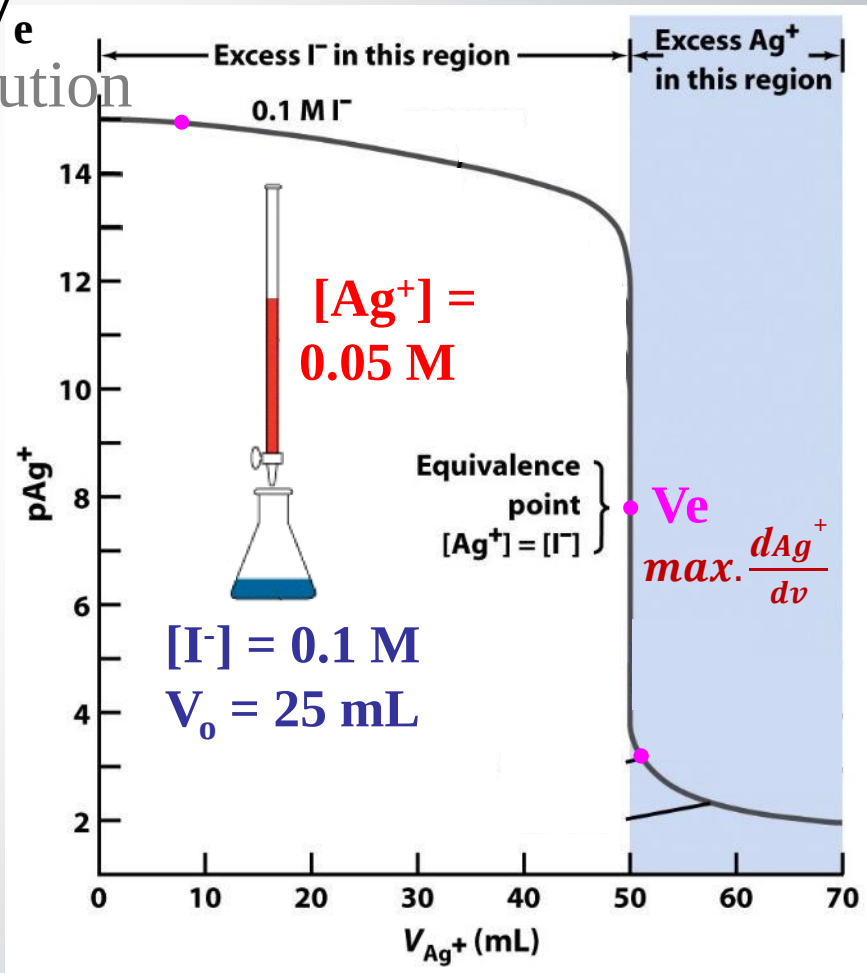
$$p[Ag^+]_f = -\log\sqrt{K_{sp}} = 8.04$$

3. After equivalence point, V > V_e

$p[Ag^+]_f$ is determined by excess Ag^+

$$[Ag^+]_f = \frac{[Ag^+]V - [I^-]V_o}{V_o + V}$$

$p[Ag^+]_f = 2.89 \ (V = 52\ mL)$



Effect of reagent concentration and K_{sp} on titration accuracy

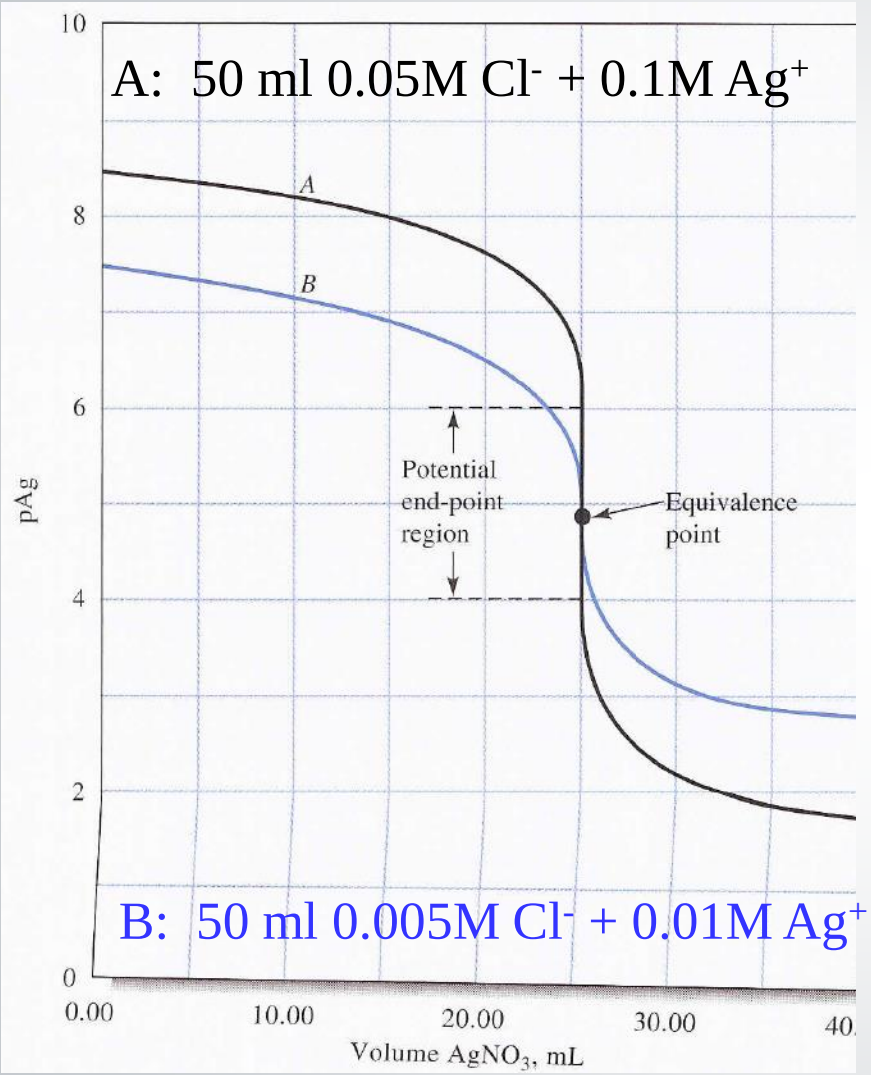
- Which titration is more accurate:

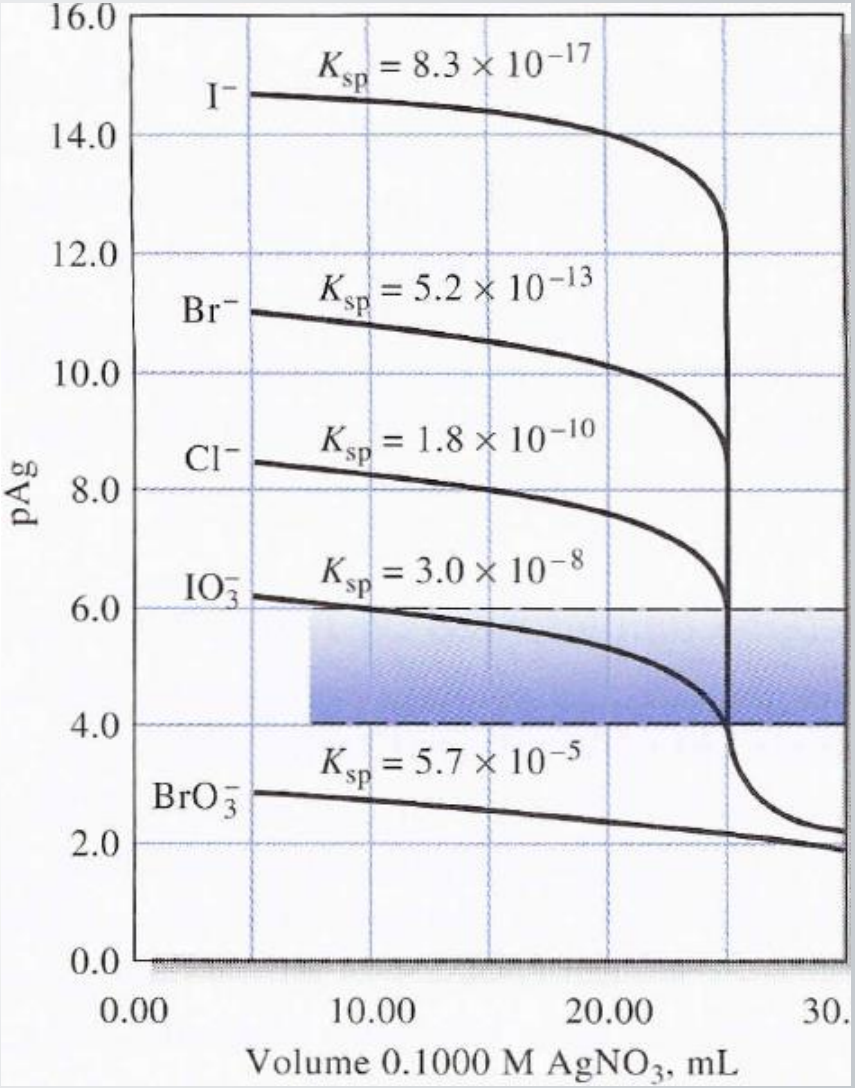
  High or low reagent concentrations?

- Which titration is more accurate:

  Ag^+ with I^- or with Cl^-?

Effect of [C] & K_{sp} on titration error





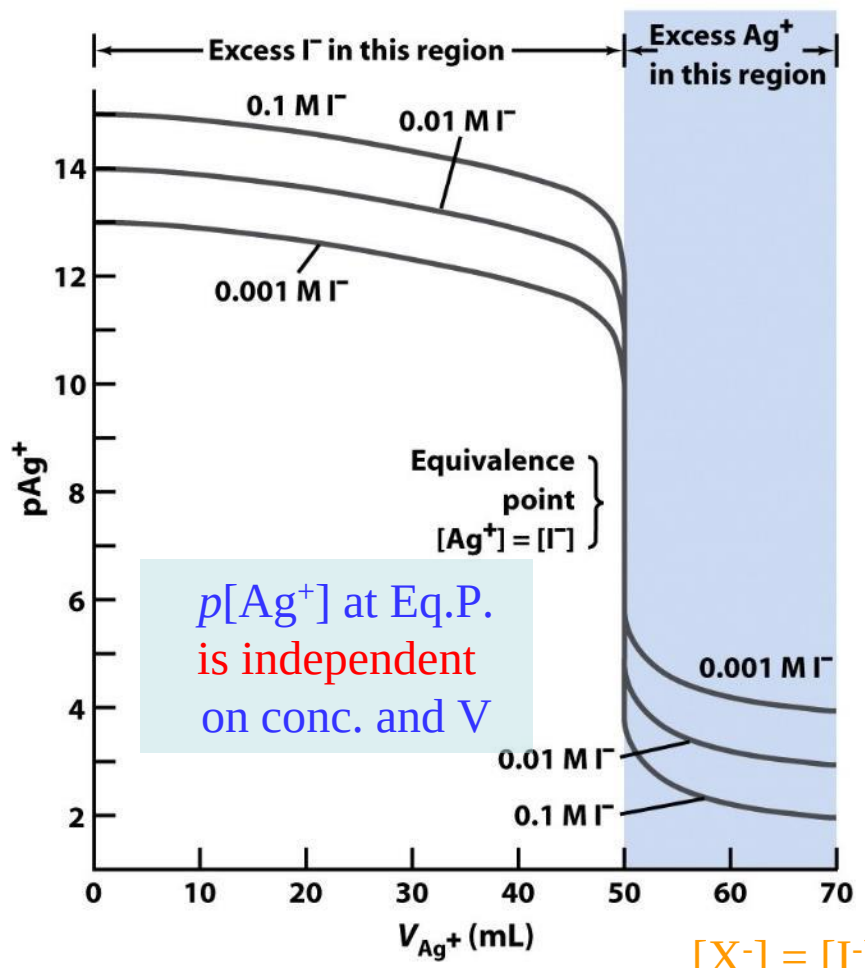
The [C] of reactants are higher, the titration error will be less

The K_{sp} of the precipitant is smaller, the titration error will be less

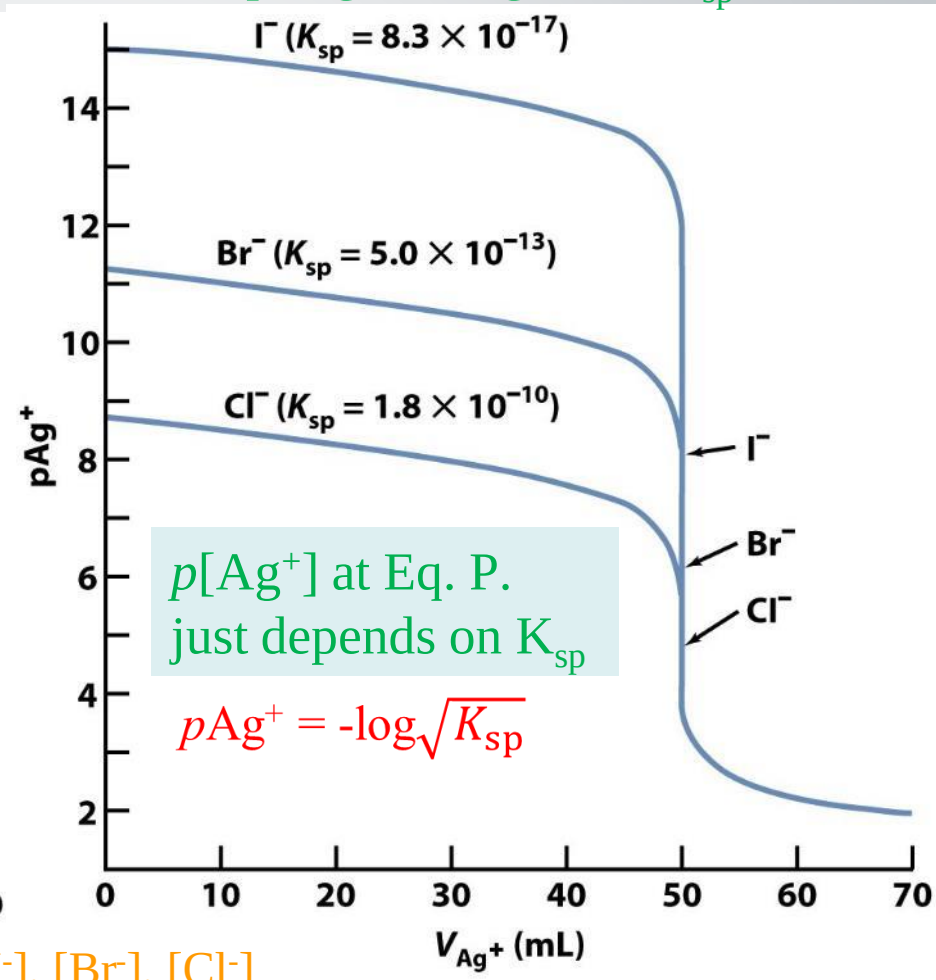
Effect of [C] & K_{sp} on titration error

Initial $p[Ag^+]$ is higher when $[I^-]$ is higher

Initial $p[Ag^+]$ is higher as K_{sp} is lower



$p[Ag^+]$ at Eq.P. is independent on conc. and V



$p[Ag^+]$ at Eq. P. just depends on K_{sp}

$pAg^+ = -\log\sqrt{K_{sp}}$

$[X^-] = [I^-], [Br^-], [Cl^-]$

Final $p[Ag^+]$ is lower as $[Ag^+]$ is higher

$$p[Ag^+] = -\log[Ag^+] = -\log\frac{K_{sp}}{[X^-]}$$

Final $p[Ag^+]$ is depend on $[Ag^+]$, which is independent on K_{sp}

Effect of reagent concentration and K_{sp} on titration error

- Which titration is less error (more accurate):

High or low reagent concentrations? High [C]

- Which titration is less error (more accurate):

Titration of I^- or Cl^- with Ag^+?

Because $K_{sp}(AgI) < K_{sp}(AgCl)$

End point detection

(a) Indicators methods: follow color change

(b) Optical methods:
Follow absorption/scattering of light

(c) Potentiometric methods: follow $p[Ag^+]$

End point detection

(a) Indicators methods based on color change

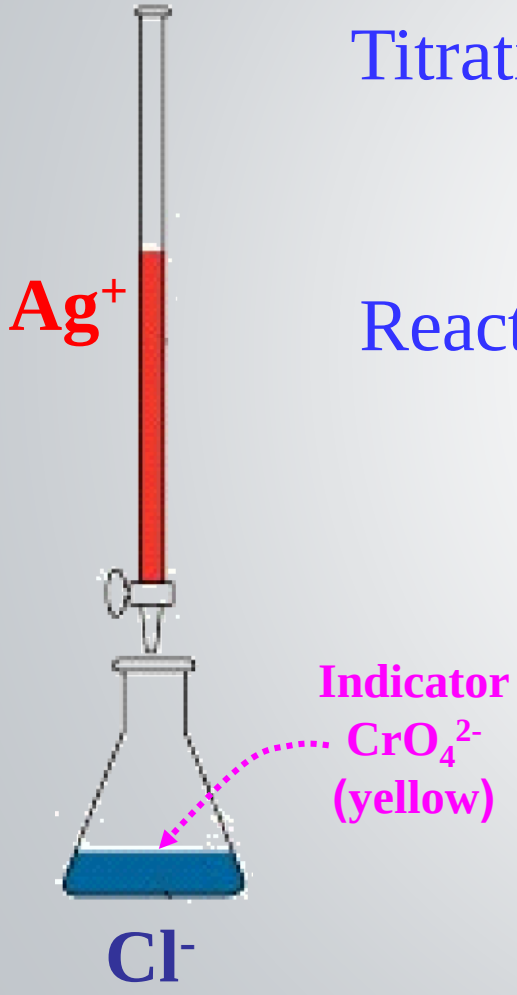
① Mohr titration

② Volhard titration

③ Fajans Titration

(a) Indicators methods based on color

① Mohr titration

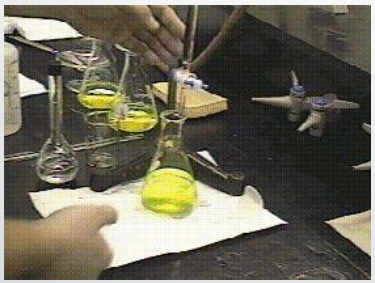


Titration reaction:

$$Ag^+ + Cl^- \rightarrow AgCl\downarrow$$

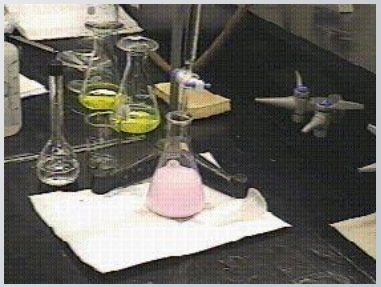
Reaction at e.p

$$2Ag^+ + CrO_4^{2-} \rightarrow \mathbf{Ag_2CrO_4}\downarrow$$



Starting point
yellow

During titration
milky white (AgCl)



End point
pink

Ensuring that the indicator precipitates at eq. p.

Ksp (AgCl) = 1.8×10^{-10}

Ksp (AgBr) = 5.0×10^{-13}

Ksp (AgI) = 8.3×10^{-17}

Ksp (Ag_2CrO_4) = 1.2×10^{-12}

What should be the indicator's concentration so that it precipitates at eq. p.?

At eq. p.: $[Ag^+] = \sqrt{Ksp\ (AgCl)} = 1.3 \cdot 10^{-5}$

$2Ag^+ + CrO_4^{2-} \rightarrow \mathbf{Ag_2CrO_4}\downarrow$ $\quad Ksp\ (Ag_2CrO_4) = [Ag^+]^2\ [CrO_4^{2-}]$

$1.2 \times 10^{-12} = (1.3 \times 10^{-5})^2\ [CrO_4^{2-}]$

$[CrO_4^{-2}] = 0.0071\ M \approx 0.01\ M$

If concentration of chromate is higher:
the pink color appears **before** eq. point
If concentration of the chromate is lower:
the pink color appears **after** the eq. point

Difficulties

1. A large quantity of $Ag_2CrO_4\downarrow$ is needed for observing the pink color. (Partial correction: titration without Cl^-)

2. K_{sp} depends on temperature (differently for each compound). The standard procedure is for room temp.

3. Accuracy depends on indicator's concentration.

4. *p*H ≈ 8 must be maintained in the solution.

   At more acidic *p*H the indicator provides $HCrO_4^-$

   At more basic *p*H, AgOH precipitates

(a) Indicators methods based on color

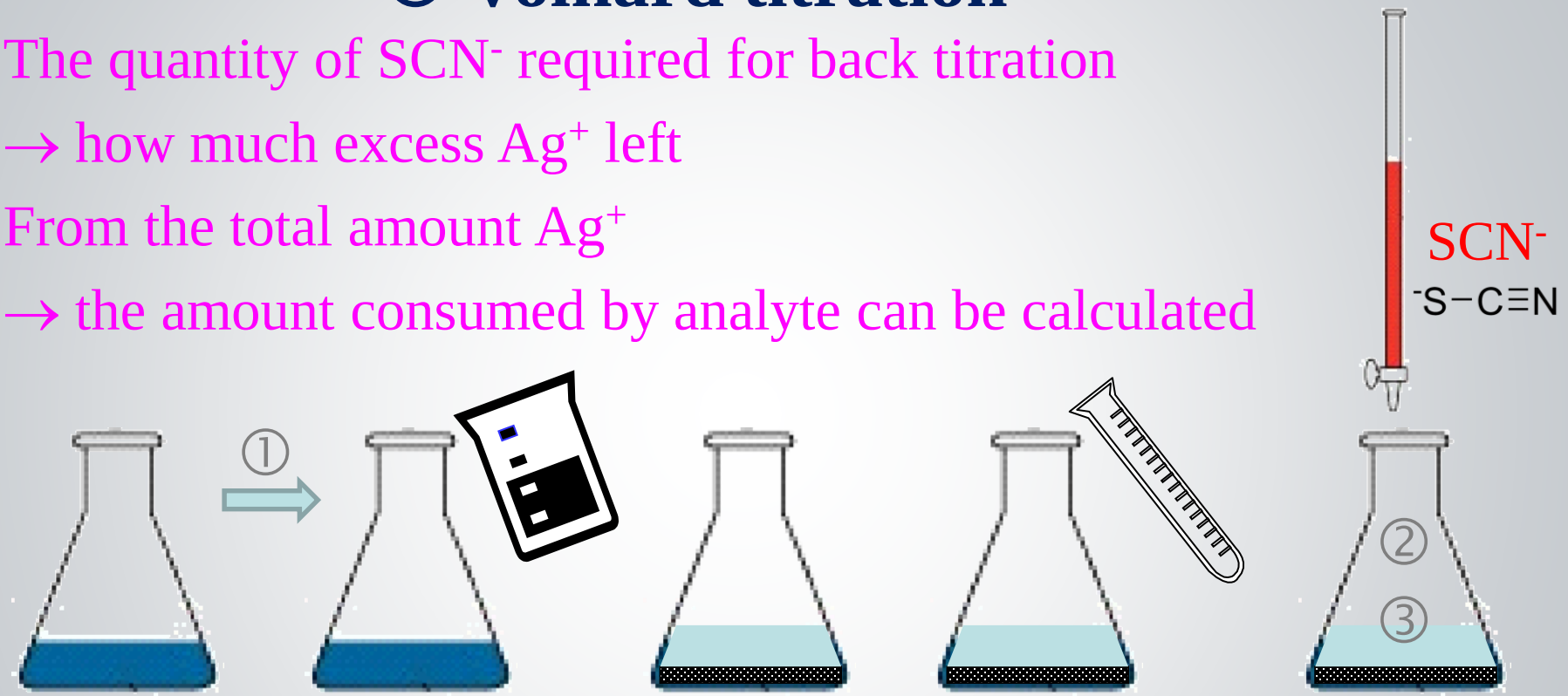
② Volhard titration

The quantity of SCN^- required for back titration

→ how much excess Ag^+ left

From the total amount Ag^+

→ the amount consumed by analyte can be calculated



Cl^-, I^-, Br^-
Analyte

Add known quantity of Ag^+ **in excess**

Precipitate + Ag^+ excess

Add Indicator of Fe^{3+}

Titrate excess Ag^+ with SCN^-

In Volhard titration, **<u>a back titration is necessary</u>**.

Some concepts you need to know

Direct Titration: add titrant to analyte until the reaction is complete.

Back Titration: add known excess of one standard reagent to the analyte, and then titrate the excess of reagent with a second standard.

Volumetric analysis: procedures in which we measure the volume of reagent needed to react with analyte.

Reactions in Volhard titrations

① $Ag^+ + Cl^- \rightarrow AgCl\downarrow$

In acidic solution (0.2 M HNO_3)

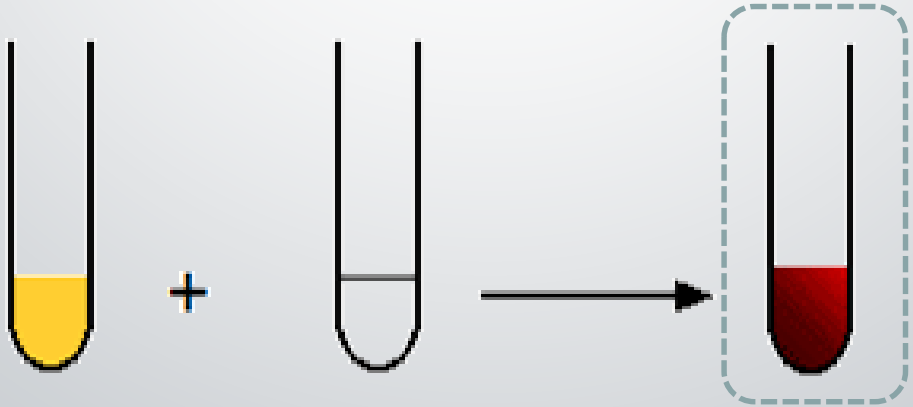
Titration reaction:

② $Ag^+ + SCN^- \rightarrow AgSCN\downarrow$

When Ag^+ has been consumed, the next drop of SCN^- reacts with Fe^{3+} to produce the red complex.

Reaction at end point:

③ $Fe^{3+} + SCN^- \rightarrow \mathbf{FeSCN^{2+}}$



The appearance of red color is the end point

Problem:

AgCl is more soluble than AgSCN. So, quantity of SCN^- becomes larger due to the follwing reaction:

$$AgCl\downarrow + SCN^- \leftrightarrows AgSCN\downarrow + Cl^-$$

K_{sp} (AgSCN) = 1.03×10^{-12}

K_{sp} (AgCl) = 1.8×10^{-10}

K_{sp} (AgBr) = 5.0×10^{-13}

K_{sp} (AgI) = 8.3×10^{-17}

AgBr and AgI do not haven such problem.

Solutions:

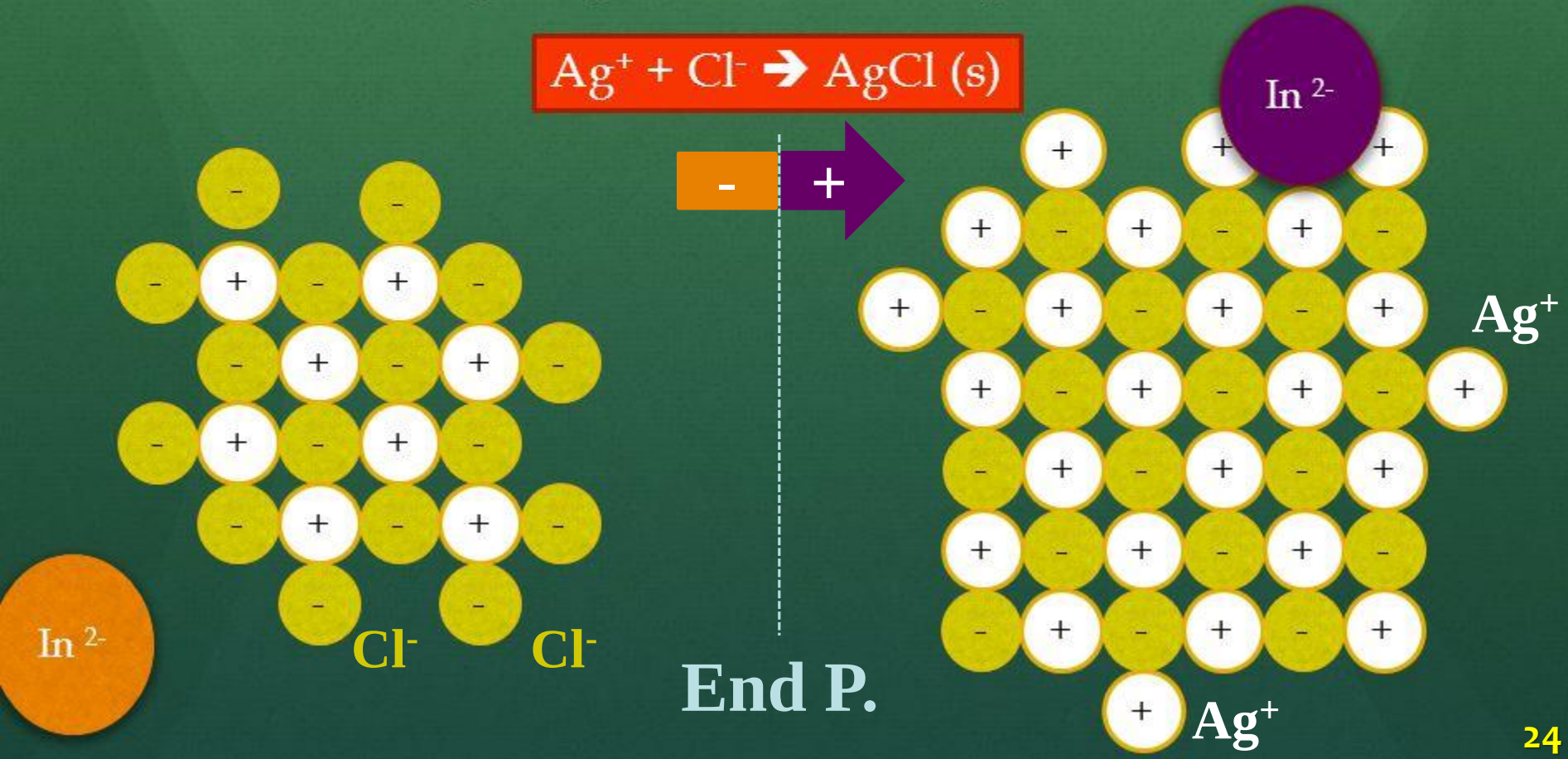
- Filtering the AgCl precipitate and titrate Ag^+ in the filtrate.
- Adding nitrobenzene that coats/isolate the AgCl precipitates.

(a) Indicators methods based on color

③ Fajans Titration

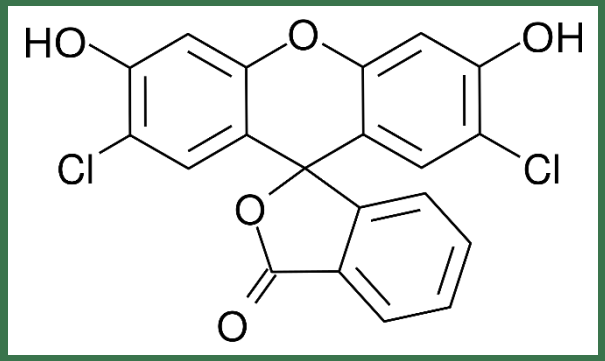
Fajans titration – involves the adsorption of a colored indicator on the precipitate at the end point.

$Ag^+ + Cl^- \rightarrow AgCl\ (s)$

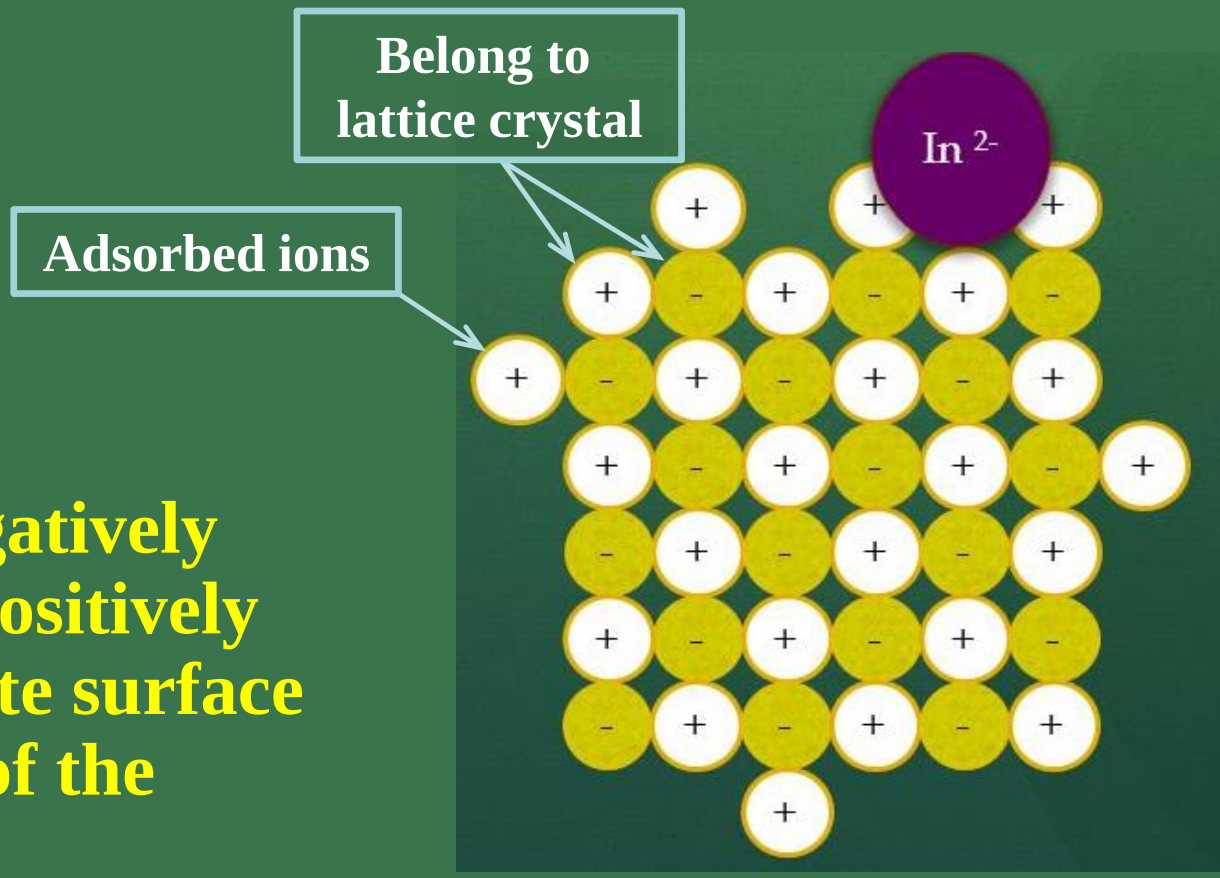


③ Fajans Titration

Absorption indicator are anionic dyes.



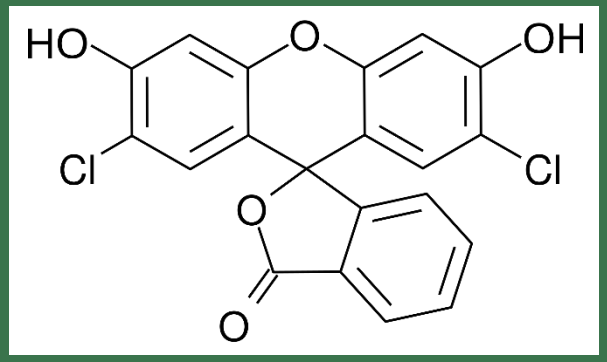
Dichlorofluorescein (DCF)



Adsorption of negatively charged dye on positively charged precipitate surface change the color of the indicator

③ Fajans Titration

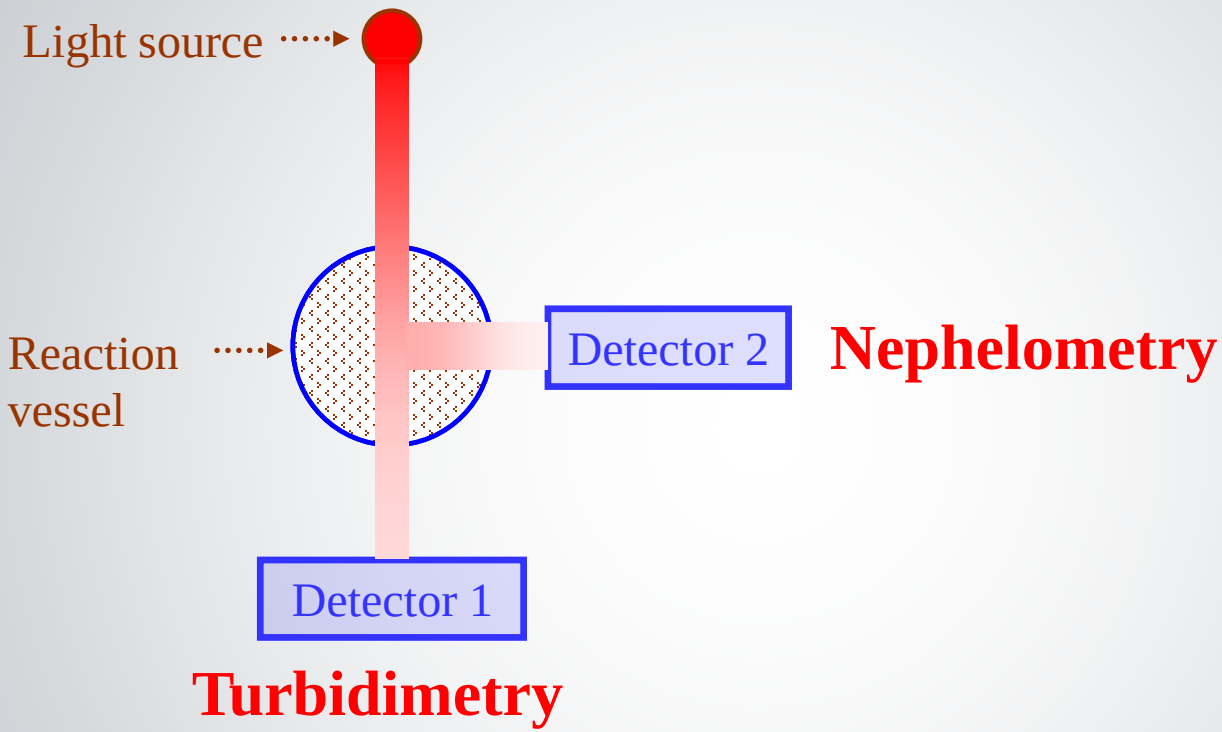
Absorption indicator are anionic dyes.

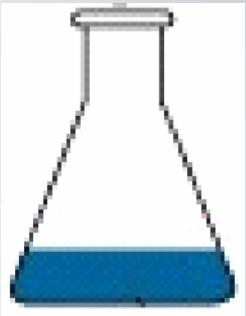


Dichlorofluorescein (DCF)

From greenish yellow to pink

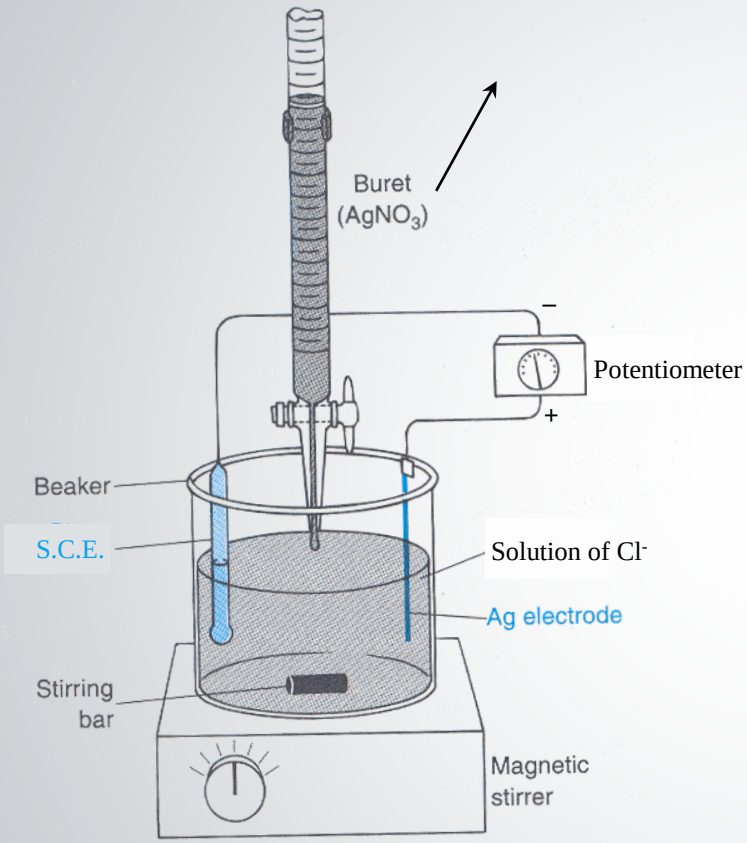
(b) Optical methods





- Very sensitive, can be used at low concentrations.
- Improvement can be achieved using spectroscopy.
  (the spectrum depends on particle size)

(c) Potentiometric precipitation titration



100 mL 0.1 M NaCl was titrated with 0.1 M $AgNO_3$ ($K_{sp} = 1.8\times10^{-10}$)

The voltage of the cell was monitored

Calculate the voltage after addition of a) 65 mL, b) 100 mL and c) 135 mL of $AgNO_3$

Apparatus for measuring titration curves. The silver electrode responds to the change of $[Ag^+]$, Saturate calomel electrode (S.C.E.) provides a reference potential

Nernst equation

For the reaction: **aA + bB ⇌ cC + dD**

Nernst eq. provides the cell potential:

$$E = E^{o} - \frac{RT}{nF} \ln \frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}}$$

E^{o} is the cell's standard reduction potential (the potential when all concentrations are 1M)

n is the number of passing electrons

Example:

$$Cd(s) \rightleftharpoons Cd^{2+} + 2e^- \qquad E^\circ = 0.402\ V$$

$$2Ag^+ + 2e^- \rightleftharpoons 2Ag(s) \qquad E^\circ = 0.799\ V$$

$$Cd(s) + 2Ag^+ \rightleftharpoons Cd^{2+} + 2Ag(s) \qquad E^\circ = 1.201\ V$$

$$E = E^\circ - \frac{0.059\,16}{2} \log \frac{[Cd^{2+}]}{[Ag^+]^2}$$

$$\frac{RT}{nF}$$

Converting the natural log into the base 10 log and T = 298.15 K,
R = 8.314 J/(K·mol), Faraday constant = 9.649×10^4 C/mol ($F = e \cdot N_A$)

Metal indicator electrodes

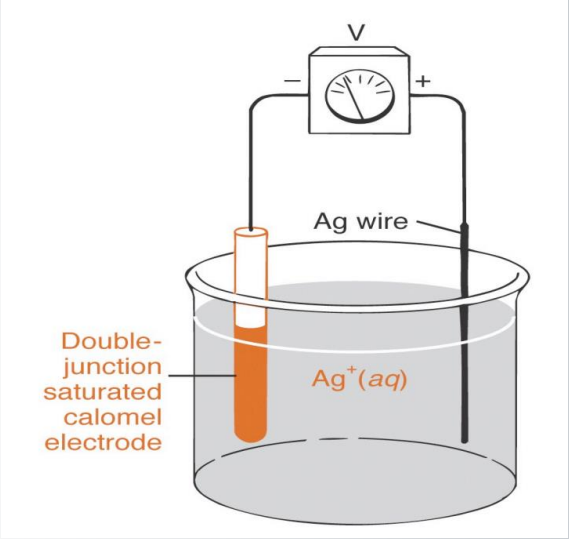
For example: use of Ag and calomel electrodes to measure $[Ag^+]$

Reaction at the Ag indicator electrode:

$Ag^+ + e \rightleftharpoons Ag\ (s) \quad E_+^o = 0.799\ V$

The reference electrode of S.C.E.:

$HgCl_2\ (s) + 2e \rightleftharpoons Hg\ (l) + 2Cl^- \quad E_- = 0.241\ V$



The Nernst equation for the entire cell is:

$$E = E_+ - E_- = \underbrace{\left(0.799 - 0.05916 \log \frac{1}{[Ag^+]}\right)}_{\text{Potential of Ag | Ag}^+\text{ indicator electrode}} - \underbrace{0.241}_{\text{Potential of S.C.E Reference electrode}}$$

$$E = 0.558 + 0.05916 \log [Ag^+]$$

(c) Potentiometric precipitation titration

a) At 65 mL, 65% $AgNO_3$ has precipitated and 35% remains:

$$[Cl^-] = 35\% \times 0.1\ M \times \frac{100\ mL}{165\ mL} = 0.0212\ M$$

$$[Ag^+] = \frac{Ksp}{[Cl^-]} = 8.5 \times 10^{-9}\ M$$

$$E = 0.558 + 0.05916 \log [Ag^+] = 0.081\ V$$

b) At 100 mL, the Ag+ comes from the dissolution of $AgNO_3$:

$$[Ag^+] = \sqrt{k_{sp}} = 1.34 \times 10^{-5}\ M$$

$$E = 0.558 + 0.05916 \log [Ag^+] = 0.2697\ V$$

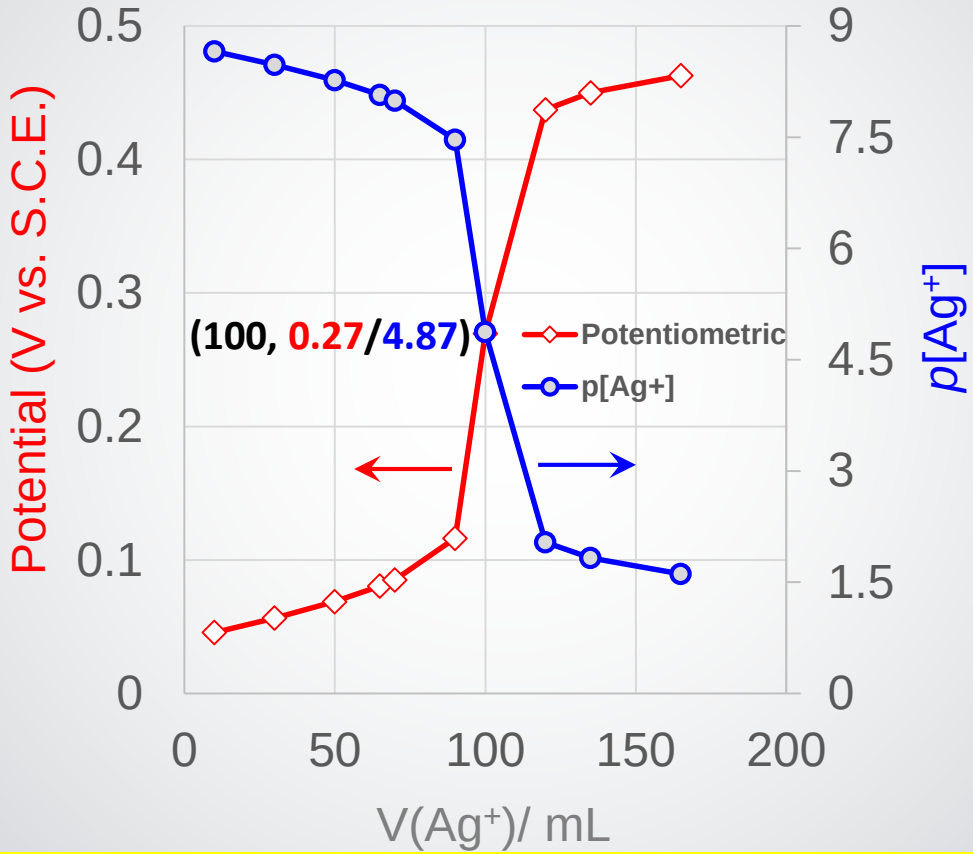
c) At 135 mL, an excess of 35 mL of $AgNO_3$ left:

$$[Ag^+] = \frac{3.5\ mmol}{235\ mL} = 0.0149\ M$$

$$E = 0.558 + 0.05916 \log [Ag^+] = 0.450\ V$$

(c) Potentiometric precipitation titration

100 mL 0.1 M NaCl was titrated with 0.1 M $AgNO_3$ ($K_{sp} = 1.8\times10^{-10}$)



$$E\uparrow = 0.558 - 0.05916\ p[Ag^+]\downarrow$$

Precipitate titration monitoring with two different methods